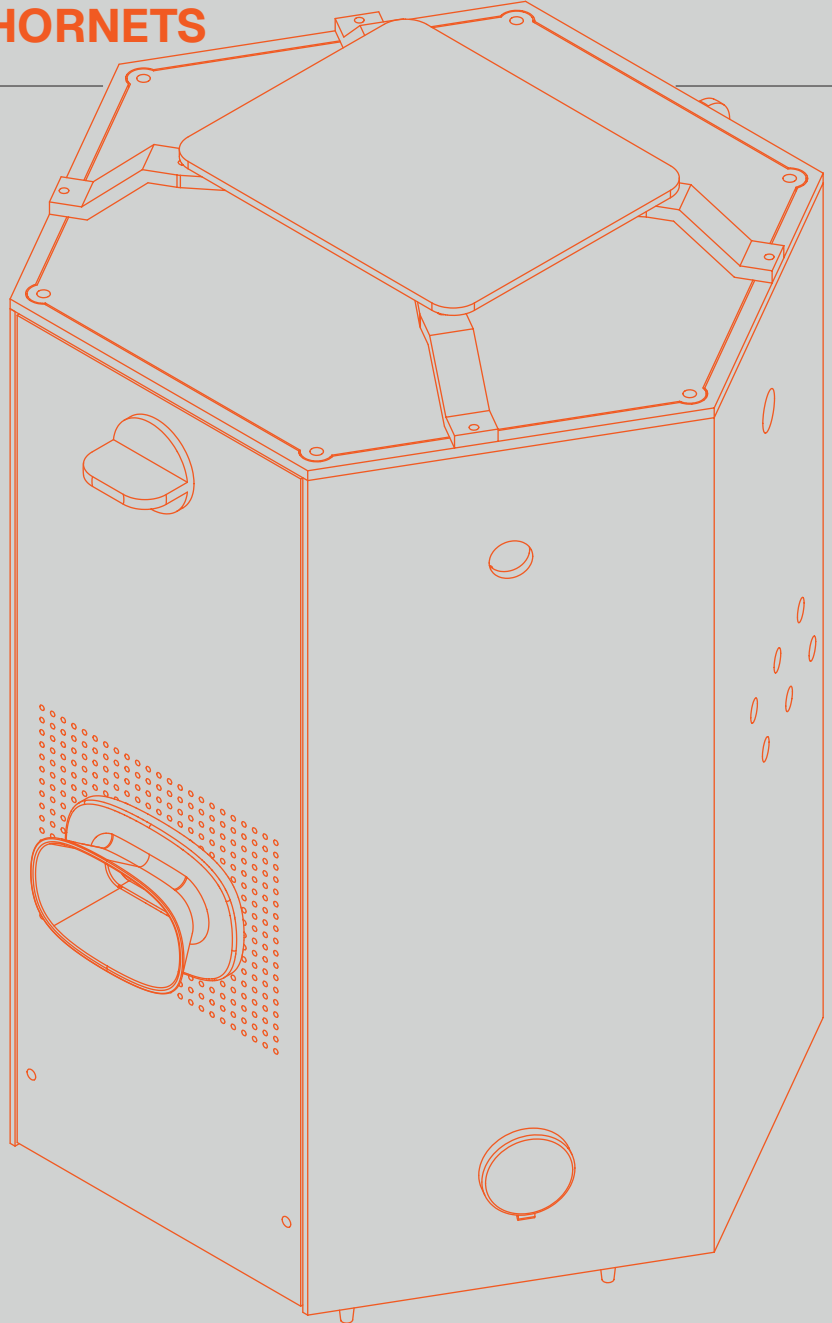


Prototype Field Test Manual

PLUG & PLAY AI DETECTION SYSTEM FOR VESPA VELUTINA HORNETS



Version: Prototype Test Edition

Status: Working proof of concept

For: Selected test users, project partners
and early field evaluators

Project: Vespa Smart Trap

Contact: ah-val@imkersleiden.nl

Repository: [github.com/aggeritsen/
VespaSmartTrap-ONE](https://github.com/aggeritsen/VespaSmartTrap-ONE)

Thank you for being part of the first VespaSmartTrap-ONE test group.

This prototype is an important step toward a selective, affordable and open-source smart trap for the Asian hornet. Your role is not only to test whether the device works, but also to help us understand how it behaves in real conditions: near hives, in gardens, in changing light, with changing bait, and with real insects passing through the trap.

The VST-ONE is not yet the final product. It is a test platform. Some parts may still feel technical. Some procedures may still require manual steps. That is exactly why this test matters.

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01 | Welcome to the field test

Your observations will help improve:

- the reliability of the AI detection
- the physical trap design
- the reset and maintenance procedure
- the user instructions
- the safety and usability of future versions
- the open-source documentation

02 | How the trap works

Three main functions:

Attract

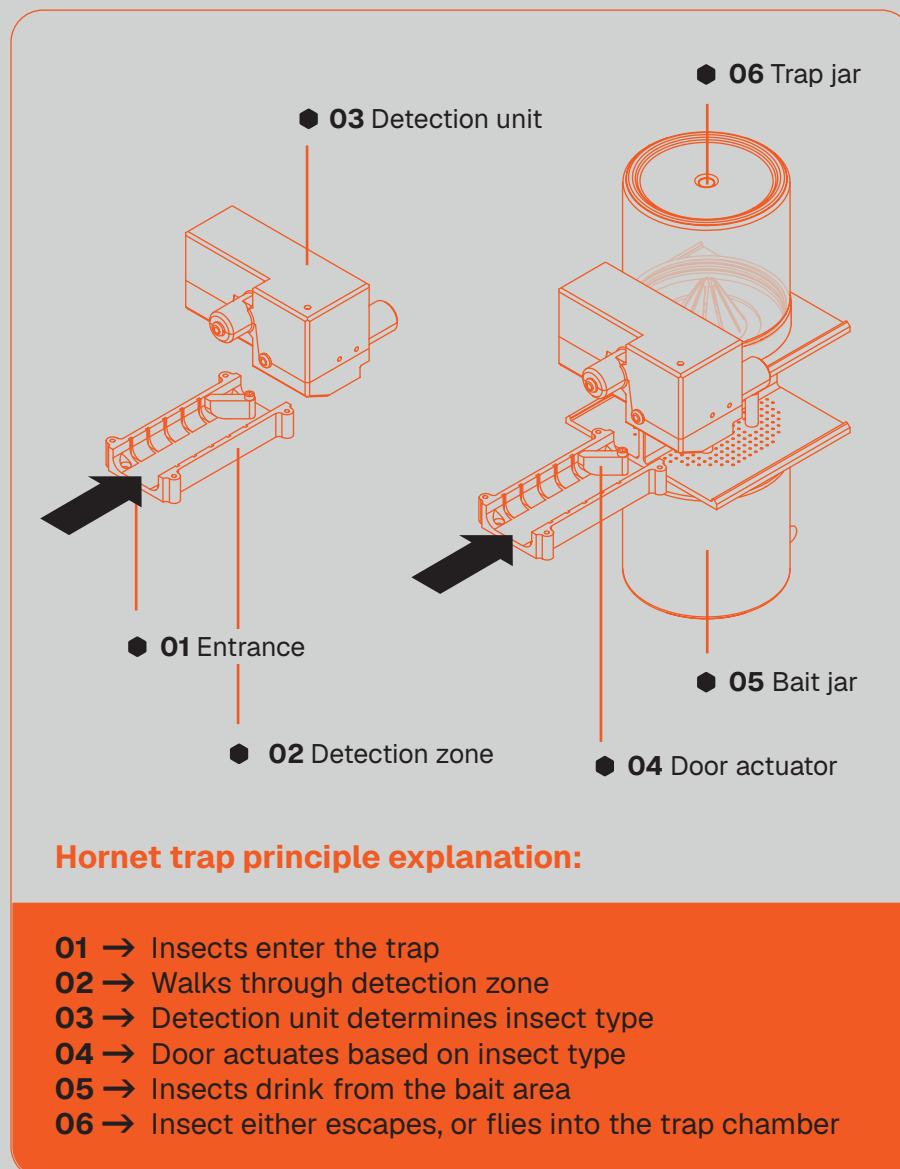
The bait area attracts insects into the observation zone.

Detect

A camera observes the insect. The AI module compares the image with its trained model.

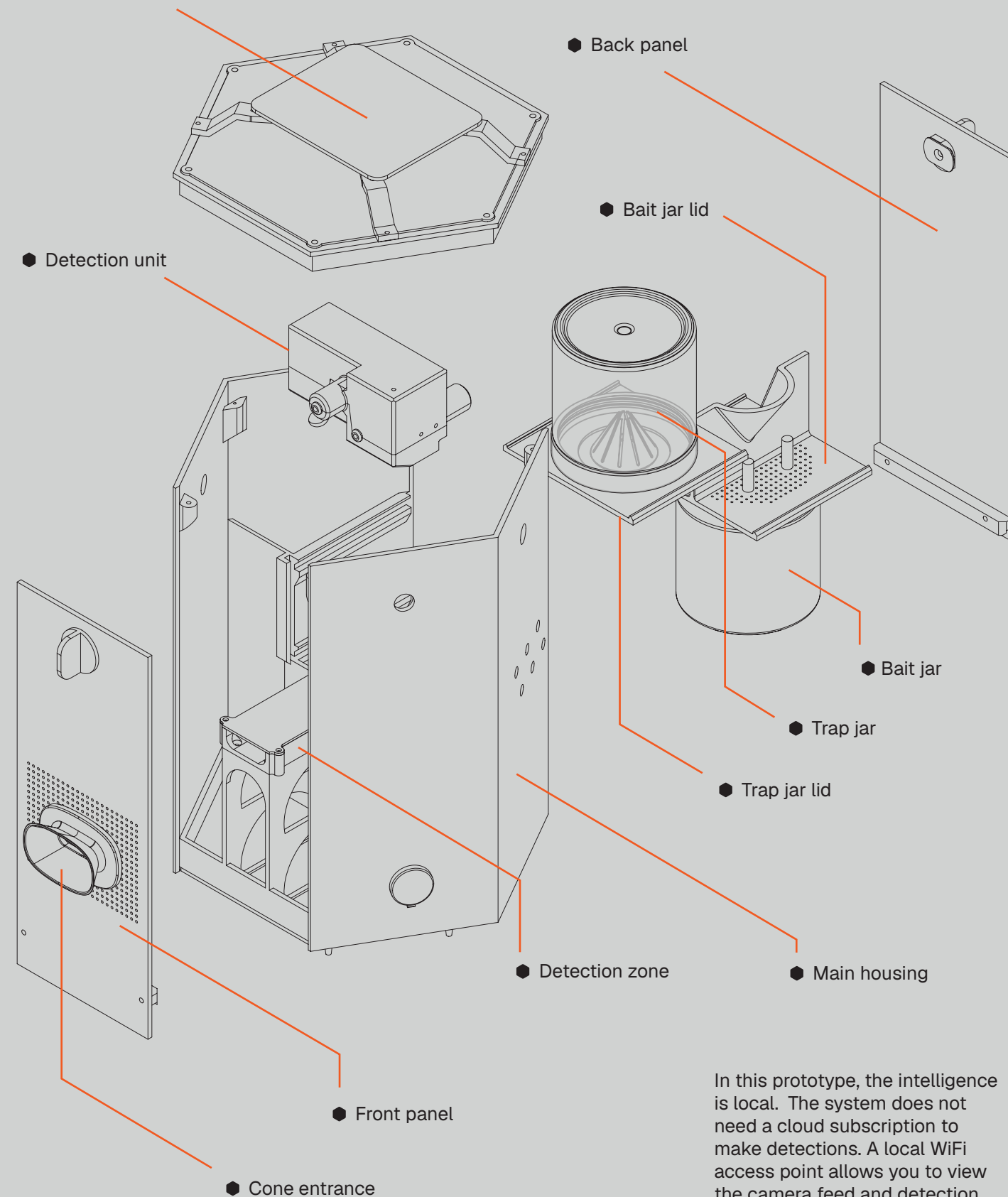
Actuate

If the system recognizes the target species above the configured confidence threshold, the controller sends a signal to the connected output.



Main elements:

- Top including solar panel



In this prototype, the intelligence is local. The system does not need a cloud subscription to make detections. A local WiFi access point allows you to view the camera feed and detection result through a browser.

03 | Before you start

In the box you will find:

- VespaSmartTrap-BASE prototype trap
- USB-C power cable
- USB-C reset plug
- Premixed bait
- Test log sheet

You will need:

- A stable outdoor location

IMPORTANT

Do not modify wiring, firmware, or mechanical parts unless the project team asks you to. This prototype is intended to be tested as supplied.

04 | Set up overview

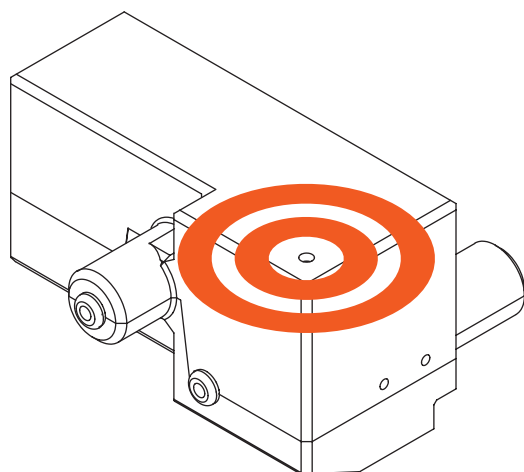
Field intelligence, built into the trap.

VespaSmartTrap-ONE uses local AI vision to recognize target species before the mechanism responds.

The next steps will guide you through the full process: setting up the trap, cleaning it, and resetting it for the next use.

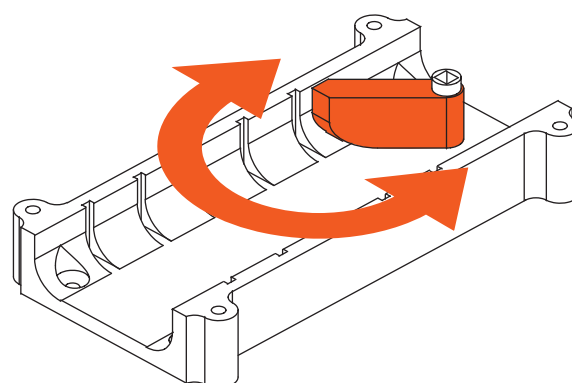
01 | Startup sequence indicators

+ INDICATOR 01



A green LED on top of the detection unit should illuminate.

+ INDICATOR 02

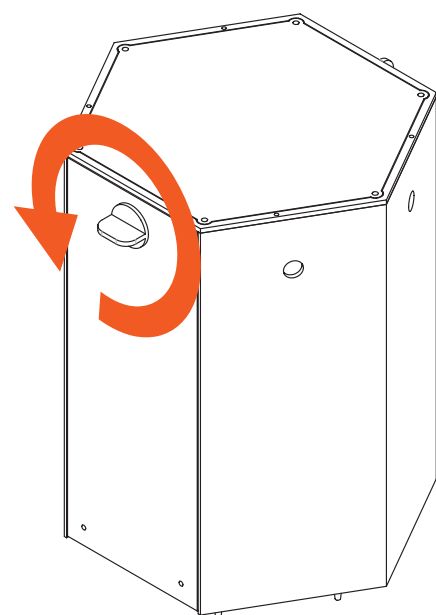


The door should automatically open and close once.

These indicate that the startup sequence has been completed successfully..
if not please reach out to ...

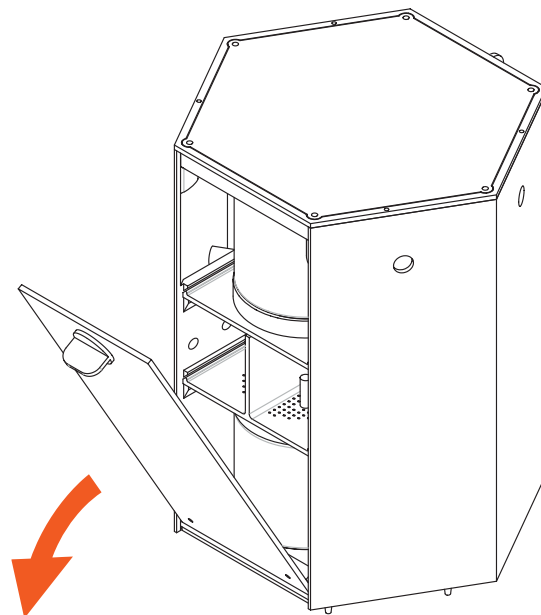
02 | Remove backpanel

+ STEP 01



Twist the button counter clockwise to unlock the back panel.

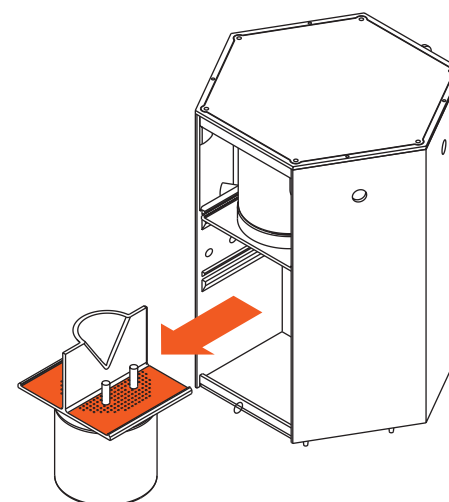
+ STEP 02



Hinge the panel outward, then remove it.

03 | Replacing the bait

+ STEP 01



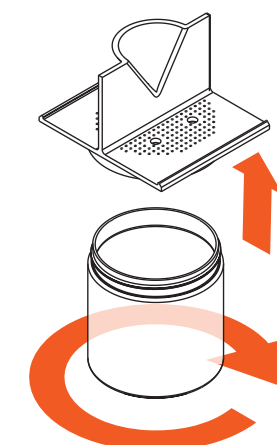
Slide the bait jar and lid holder outward by grabbing the plastic part (highlighted orange).

+ STEP 02



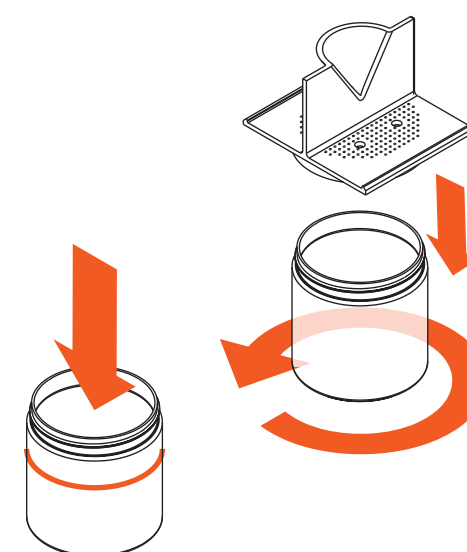
Remove the cotton sticks (4x) by sliding them upward.

+ STEP 03



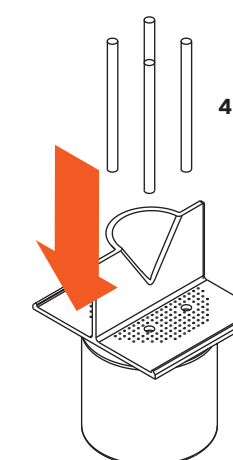
Twist the bait jar to loosen and remove it from the holder.

+ STEP 04



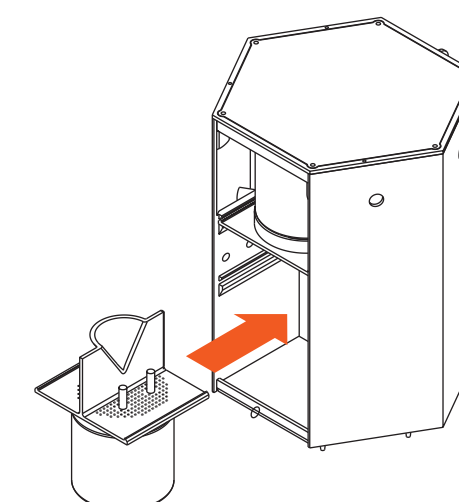
Fill the bait jar with provided bait fluid. Don't exceed fill line. Twist the bait jar lid back onto the holder.

+ STEP 05



Put new cotton sticks (4x) in the bait jar by sliding them through the holes.

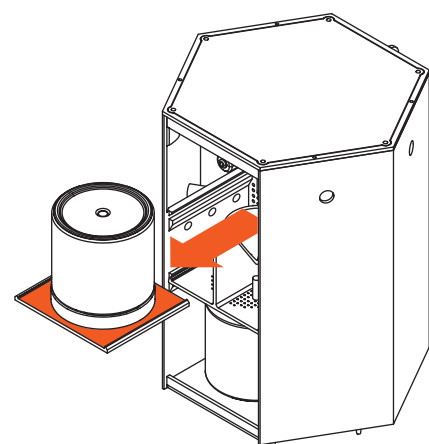
+ STEP 06



Slide bait holder back into the product and close the backpanel again.

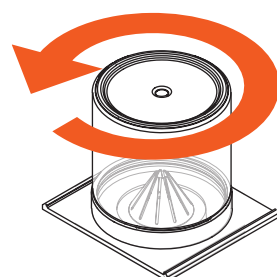
04 | Removing a trapped hornet

+ STEP 01



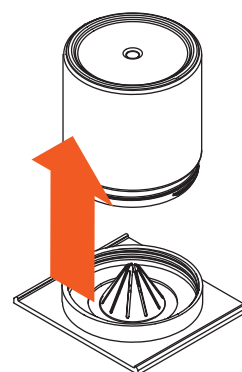
Slide the trap jar holder outward by grabbing the trap jar lid and pulling it out (highlighted orange).

+ STEP 02



Twist the trap jar to loosen.

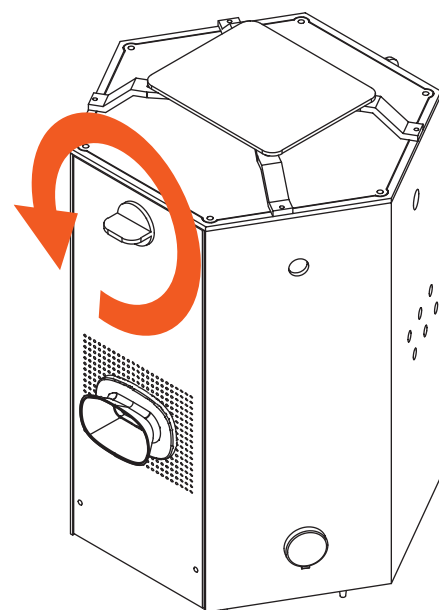
+ STEP 03



Remove it from the trap jar lid.

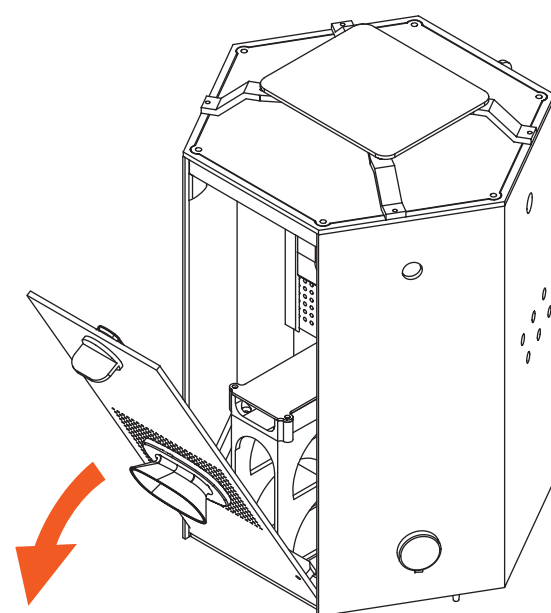
05 | Remove front panel for healh check

+ STEP 01



Twist the button counter clockwise to unlock the front panel.

+ STEP 02

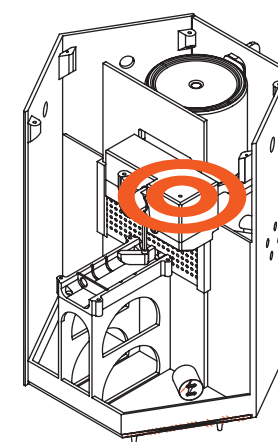


Hinge the panel outward, then remove it.

The health check can be performed without opening the product, by looking through the clear panels. If desired, the front panel can be removed to increase visibility, or for cleaning purposes

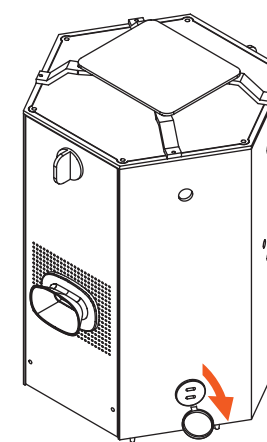
06 | Setup / health check

+ STEP 01



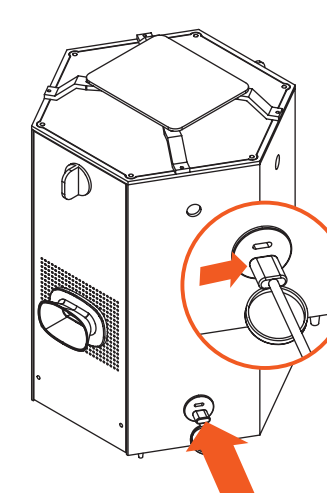
Check if the system is running. The device is always be powered by the internal battery. You can check if its running by checking the LED, which should blink red every minute and close once.

+ STEP 02



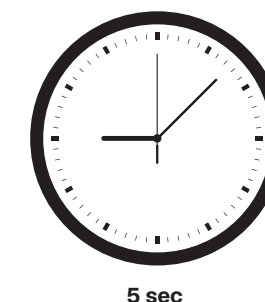
Reset the device | step 01
Open the waterproof cap at the bottom of the device.

+ STEP 03



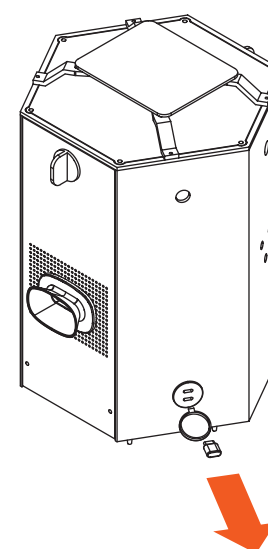
Reset the device | step 02
Plug in the USB-C reset key in the **bottom usb-C connector**.

+ STEP 04



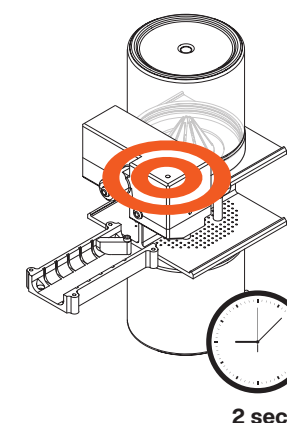
Reset the device | step 03
Wait for 5 seconds

+ STEP 05



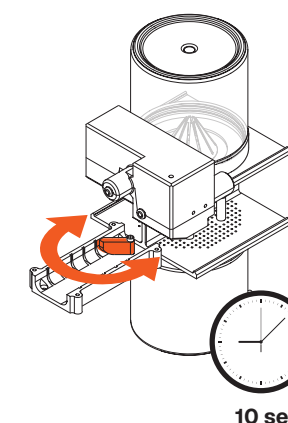
Reset the device | step 04
Remove the USB-C reset key from the bottom usb-C connector

+ STEP 06



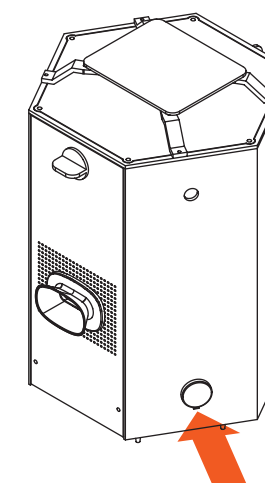
Reset the device | step 05
2 seconds after removing the connector from step 4, LED should blink red.

+ STEP 07



Reset the device | step 06
The door should switch position, Then, after another 10 seconds, the door should switch back

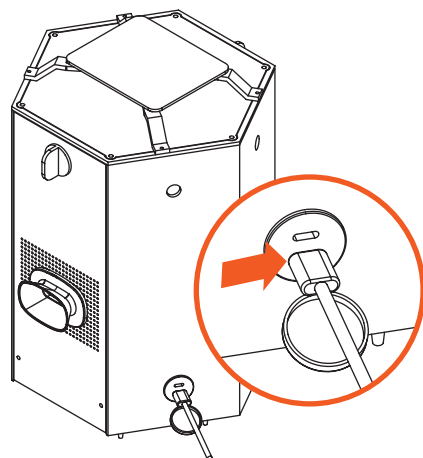
+ STEP 08



Reset the device | step 07
Fix the waterproof cap back onto the connection plug

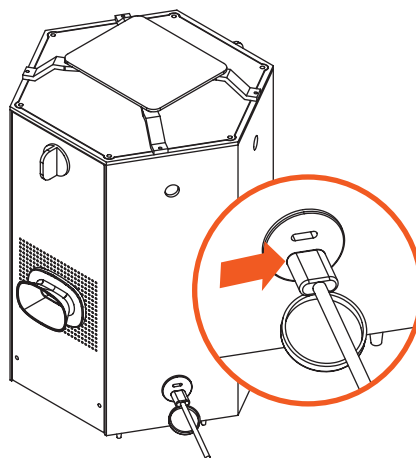
07 | Charging and data connection

STEP 01



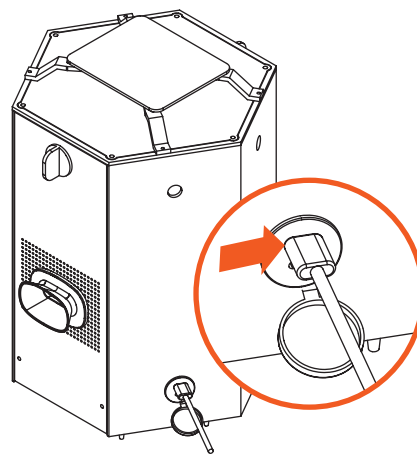
To charge the device, plug the provided USB-C cable in the **bottom connector** and connect it to a power source unit should illuminate.

STEP 02



To read out the firmware log, or flash new firmware, plug the provided USB-C cable in the **bottom connector** and connect it to a laptop.

STEP 03



To read out the Groove vision video directly, plug the provided USB-C cable in the **top connector** and connect it to a laptop.

08 | Daily test log

In order to get the right insights from the test, the following tasks need to be performed daily. Use the provided sheet to log.

- 01** → Check whether the device is running
- 02** → Perform a health check (refer to page x)
- 03** → Inspect the bait area; is there still plenty of bait in the pot?
If there is less than 30% inside, please refill the bait according to instructions on page x
- 04** → Inspect the trap container
If a hornet is present in the chamber, you can let it stay there until it dies?
Are there other animals present inside the chamber
- 05** → Inspect the cotton sticks
Are they still sticking out into the chamber?
If not, replace the cotton sticks

To get reliable insights from the test,
complete the tasks in the table on the next page every day.

- Use one row per day: tick each task once done, note how many hornets you saw, and add anything notable in the comments.

[illegible]

What each column means:

- 01 → DEVICE RUNNING:** Check whether the device is running.
- 02 → HEALTH CHECK:** Perform a health check (see manual).
- 03 → BAIT >30%:** Inspect bait area : tick if pot is still over 30% full.
- 04 → BAIT REFILLED:** If bait is below 30%, refill it (see manual) and tick.
- 05 → TRAP CHECKED:** Inspect the trap container : tick when checked.
A trapped hornet may be left until it dies; note other findings under Comments.
- 06 → OTHER ANIMALS:** Tick if other (non-hornet) animals were present in the chamber.
- 07 → COTTON STICKS OK?:** Cotton sticks still poking into the chamber? Tick if OK; if not, replace them.
(see manual)

09 | Field notes

Use this space to write down anything you notice during testing: trap location, weather conditions, insect behaviour, bait condition, or changes around the trap. Small observations can be useful later when comparing test results.

10 | Feedback & ideas

Use this space to share your experience with the prototype. Note what worked well, what was unclear, what could be improved, and any ideas that may help make the next version better.

QUESTIONS, ISSUES OR TEST FEEDBACK?

Email

ah-val@imkersleiden.nl

GitHub repository

github.com/aggeritsen/VespaSmartTrap-ONE